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Potential skill of continental-scale, medium-range ensemble streamflow forecasts for flood prediction in South America

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ABSTRACT

Recent scientific and computational developments have pushed hydrological forecasting systems up to continental and global scales. In this study, we evaluate the potential skill of ensemble streamflow forecasts for flood prediction in South America (SA), with a closer look on some of the large basins of the continent. Medium-range precipitation forecasts from the ECMWF Ensemble Prediction System were used to force a continental-scale, fully coupled hydrologic-hydrodynamic model to produce streamflow forecasts up to 15 days in advance. Forecasts were evaluated over a 4-year period against a reference simulation, derived from the hydrological model forced with a gridded precipitation dataset that merges data from multiple sources (MSWEP). To provide a benchmark for assessing the potential skill, forecasts based on the Ensemble Streamflow Prediction (ESP) technique were produced by running the hydrological model with precipitation data resampled from past MSWEP observations. Overall, ECMWF-based forecasts were skillful mainly at the eastern and southeastern regions of the continent (e.g., São Francisco, Parnaíba, Upper Paraná, Tocantins-Araguaia and Uruguay basins), showing little or no added skill in large rivers draining the Andes such as the Magdalena, Orinoco, Madeira (Amazon) and right-bank tributaries of the Paraguay. In the latter basins, ECMWF-based forecasts largely overestimated the exceedance probabilities of critical thresholds due to a positive bias in the predicted precipitation (in relation to MSWEP), visible mostly during the wet season. In addition, forecasts were found to perform better than ESP in terms of both overall quality and discrimination skill at regions characterized by a warm temperate, fully-humid climate. A direct relation between forecast skill and basin size was not always valid, since the former can vary geographically along tributaries. Limitations and future developments needed for continental-scale hydrological forecasting in SA are also discussed.

1. Introduction

Floods are among the most common water-related disasters worldwide, causing several impacts including human losses, issues on public health and damages to property and infrastructure. According to EM–DAT database (Guha-Sapir et al., 2018), riverine floods alone were responsible for at least US$ 20 billion in economic losses, 38 million affected people and 1 million homeless in South America (SA) between 1980 and 2017, partly because of the rapid population growth, urban expansion and occupation of riparian zones in the last century with consequent exposition to risk (Nunes, 2011; Tucci and Bertoni, 2003; Vorosmarty et al., 2013). Recent extreme, large-scale flood events have been documented in scientific literature for Amazon region (Espinoza et al., 2014, 2013; Marengo et al., 2012b), Colombia (Hoyos et al., 2013), Southeastern Brazil (Tomasella et al., 2019) and also in international headlines (e.g., 2015 floods in Paraguay, Argentina, Uruguay and Southern Brazil) due their widespread effects and great repercussion. Besides, projected impacts of climate change point to an increase of the frequency of current 100-yr flood magnitude over much of SA (Arnell and Gosling, 2016; Hirabayashi et al., 2013). Therefore, there is a need to improve adaptation measures and preparedness through feasible and cost-effective solutions, especially in less developed regions where financial resources are often limited.

Provision of flood forecasting and early warning systems (EWSs) can

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bring considerable benefits to anticipate flood events. EWSs can trigger preparedness actions among civil defense organizations, local governments, and people in potentially affected areas, allowing them to respond timely in order to reduce emergency costs (Alfieri et al., 2018; Jacks et al., 2010; Moore et al., 2005; Rogers and Tsirkunov, 2011). To extend the lead time of forecasts issued by EWSs and consequently enable more effective actions, one solution is to force hydrological models with precipitation outputs from numerical weather prediction systems (NWPs). The ability to predict river flows with longer lead times comes, however, with the drawback that such forecasts are subject to large uncertainties. For instance, small errors on the initial conditions of NWPs may rapidly evolve — in a chaotic manner — into large discrepancies on future states of the atmosphere (Kalnay, 2003; Leutbecher and Palmer, 2008), while fine-scale processes related to rainfall such as cloud microphysics or deep convection need to be resolved with parameterization schemes due to their extreme complexity and relatively coarse resolution of numerical grids (Mesinger et al., 2012; Song et al., 2012). Similarly, hydrological models suffer from errors in input data, simplification of model equations, and deficiencies in parameter estimation, thus amplifying the uncertainty in streamflow (Velázquez et al., 2011; Zappa et al., 2011). The use of ensemble techniques, originally from the meteorology field (Kalnay, 2003), makes it possible to account for the uncertainties of distinct components of the forecasting chain (Cloke and Pappenberger, 2009; Schaake et al., 2007), where the most common approach is to force a hydrological model with multiple forecast scenarios from Ensemble Prediction Systems (EPS). By enabling a probabilistic assessment of the forecasts, such as the likelihood of occurrence of a given flood magnitude, ensemble hydrological predictions are generally more useful than deterministic ones that rely on a single forecast outcome (e.g., Boucher et al., 2012; Jaun and Ahrens, 2009; Ramos et al., 2013; Roulin, 2007; Schellekens et al., 2011; Verkade and Werner, 2011), and are recognized as the only way to achieve skillful predictions for longer lead times (Pappenberger et al., 2011).

Advances in computational power, atmospheric prediction (Buizza and Leutbecher, 2015) and large-scale hydrological/river routing modeling (Bierkens, 2015; Yamazaki et al., 2013) are encouraging ensemble flood forecasting systems to be applied not only at regional scale but also at the continental and global domains. Some operational examples are the European Flood Awareness System (EFAS) (Thielen et al., 2009), the Hydrological Ensemble Forecasting System (HEFS) (Demargne et al., 2014), the Global Flood Awareness System (GloFAS) (Alfieri et al., 2013; Emerton et al., 2018) and its derivatives (e.g., Snow et al., 2016), which are currently providing early warning information and future river conditions for the next weeks or even months ahead (see Emerton et al. (2016) for an overview about this topic). Continental to global-scale forecasting systems can foster knowledge about flood conditions beyond the catchment boundaries, increase awareness and support coordination at international level (Alfieri et al., 2013, 2014; Emerton et al., 2016; Thielen et al., 2009; Thiemig et al., 2015), as long as the massive information produced by ensemble predictions is properly communicated to end users (Ramos et al., 2010). The use of state-of-the-art atmospheric and hydrological modeling techniques have leveraged the monetary benefit of these systems (Pappenberger et al., 2015a), thus aiding to narrow the gap between research and operational activities (Emerton et al., 2016; Pagano et al., 2014).

The benefits of coupling hydrological models with atmospheric EPSs have been widely explored over the past decade, mostly in developed countries (see the review by Cloke and Pappenberger, 2009), but in SA this topic has received attention only in the last years. For instance, ensemble forecasting approaches have been increasingly tested at short- to medium range (3–15 days) in order to assess the potential of supporting early warning systems (e.g., Casagrande et al., 2017; Meller et al., 2014; Siqueira et al., 2016; Tomasella et al., 2019) and to optimize reservoir operations under multiple objectives (e.g., Fan et al., 2014, 2016b, 2015; Quedi and Fan, 2020; Raso et al., 2014; Schwanenberg et al., 2015). However, there is currently no integrated ensemble forecasting of national and transnational basins in SA from local institutions due to the complexity in simulating large South American rivers (Paiva et al., 2013a; Siqueira et al., 2018), challenges in handling multiple model runs and processing large amounts of data, and low coverage of real-time hydrometeorological monitoring for forecasting needs (Fan et al., 2014, 2016a, 2015; Salio et al., 2015; Tomasella et al., 2019). Efforts toward continental hydrologic–hydrodynamic modeling in SA (Siqueira et al., 2018) and new precipitation datasets that can be obtained in real time (Beck et al., 2017a,b) may be the starting point to overcome such limitations and to expand hydrological forecasting initiatives in the region also to broader scales.

Routinely issuing ensemble hydrological forecasts at the continental scale requires considerable institutional and technical developments, so there must be a prior knowledge about the skill of these forecasts to outline potential strengths and deficiencies. In general, the streamflow forecast skill depends on the relative importance of each source of forecast error (Greuell et al., 2019; Paiva et al., 2012; Sharma et al., 2019). It may be useful to isolate one of the major sources of uncertainty (e.g., Greuell et al., 2019; Olsson and Lindström, 2008; Pappenberger et al., 2011), for example, if one wants to assess whether the rainfall from the meteorological ensemble is able to provide skillful predictions after propagated through the hydrological system (e.g., Alfieri et al., 2013, 2014; Bartholmes et al., 2009; Thirel et al., 2008; Voisin et al., 2011). This is sometimes treated as a potential (or theoretical) skill, in which a hydrological model run is used as the “truth” to neglect the associated model biases (Candogan Yossef et al., 2017; Greuell et al., 2019; van Dijk et al., 2013). Forecast skill may be evaluated by checking attributes like the agreement of the ensemble distribution to the corresponding observations (or a historical model run) and the ability to warn about the occurrence of floods by detecting the exceedance of critical flow thresholds (Alfieri et al., 2013; Bartholmes et al., 2009), in terms of how the performance would be compared with simpler alternative systems (Pappenberger et al., 2015b). Besides, many studies evaluating hydrological forecasts over large domains or multiple basins have indicated that forecast skill has a close relationship with catchment size and geographic location (e.g., Alfieri et al., 2013, 2014; Bennett et al., 2014; Pappenberger et al., 2015b, 2011; Quedi and Fan, 2020; Thirel et al., 2008). As the continental extent of SA encompasses a wide range of climate types and hydrological conditions, understanding the spatial patterns of forecast skill becomes important to highlight regional differences and to provide insights on how the skill in a given point of interest can be linked to different areas within a same basin.

While some studies have evaluated the performance and skill of hydrological ensemble forecasts from continental scale operational systems (e.g., Alfieri et al., 2014; Bartholmes et al., 2009; Bennett et al., 2014; Brown et al., 2014), in SA this has been done only under a global perspective (e.g., Alfieri et al., 2013) and through a few number of regional studies focusing on specific basins where monitoring is a priority or data are available from hydropower companies (e.g., Fan et al., 2014, 2016a, 2016b; Siqueira et al., 2016; Tomasella et al., 2019), with few exceptions (e.g., Bischiniotis et al., 2019). Today, there are still knowledge gaps on how the forecast skill varies across different geographic locations, basins, catchment areas and climates of SA. In this context, the aim of this study is to assess the potential skill of medium-range ensemble streamflow forecasts in South America, with a particular focus on flood prediction. This paper is structured as follows. Section 2 provides a brief description about the hydroclimatic characteristics of SA. Section 3 describes the hydrological model and meteorological inputs, methods of producing ensemble streamflow forecasts, strategy of forecast skill evaluation and verification metrics. Results and discussion are presented in Sections 4 and 5, respectively. Finally, the key conclusions are presented in Section 6.

2. Hydroclimatic features of South America

South America (SA) features unique tropical, subtropical and extratropical weather and climate patterns (Satyamurty et al., 1998), with an average annual precipitation ranging from less than 1 mm yr^{-1} in the Atacama Desert (Houston, 2006) to around 10,000–13,000 mm yr^{-1} in the Chocó region in Colombia (Poveda and Mesa, 2000). In a nutshell, the atmospheric phenomena are strongly influenced by the adjacent oceans and topographic barriers such as the Brazilian Plateau and the complex, prominent Andes Cordillera, which extends along the western side of the continent between the northern and southern tips (10° N–55° S) with altitudes exceeding 4 km in some areas (Garreaud et al., 2009). Much of the precipitation over tropical and subtropical SA emerges from deep or wide convective cores that can evolve to broad stratiform regions (Romatschke and Houze, 2010), whereas deep stratiform clouds that develop along cold and warm fronts play a major role in the extratropics (Garreaud et al., 2009). Within northern SA the annual cycle of precipitation follows the meridional migration of the Intertropical Convergence Zone (ITCZ), producing a variety of unimodal and bimodal patterns (Poveda et al., 2006). Besides, the seasonality of rainfall in most areas between the equator and 25° S is connected to the South American Monsoon System (SAMS), exhibiting more than 50% of annual precipitation in austral summer (DJF) and less than 5% in austral winter (JJA) (Marengo et al., 2012a). An important feature of the regional climate is that significant amounts of moisture from tropical latitudes (i.e., Amazon) are transported to the southeastern SA throughout the year (Vera et al., 2006), which makes the latter region absent of a distinct rainy season due to the passage of cold fronts producing abundant precipitation during the winter (Vera et al., 2002). This moisture flow also contributes to the development of spatially extensive, long-lasting Mesoscale Convective Systems (MCSs) east of the Andes in regions like northern Argentina, southern Brazil and Uruguay, featuring some of the most extreme convective systems on Earth (Romatschke and Houze, 2010; Salio et al., 2007 and references therein). In southern SA precipitation is markedly asymmetric; frontal systems act in conjunction with orographic processes in the western Andean slopes, exhibiting a wet JJA in both sides of the Cordillera (it gradually changes to fully-humid south of ~35° S), but with drastic reductions of annual precipitation in Argentinean Patagonia (Garreaud et al., 2009; Viale et al., 2019).

In terms of continental hydrology, SA holds large transnational basins such as the Amazon, La Plata and Orinoco, as well as other basins with national relevance including the São Francisco, Tocantins-Araguaia, Parnaíba and Magdalena. Some large rivers like the Paraguay, the Amazon and its tributaries are highly influenced by floodplain effects, with long lasting floods that can extend from weeks to months (Paiva et al., 2013a, 2012; Paz et al., 2011). An overview of the hydroclimatic characteristics of SA is shown in Fig. 1.

3. Data and methods

3.1. MGB hydrologic-hydrodynamic model

The MGB – *Modelo Hidrológico de Grandes Bacias* (acronym of Large-Scale Hydrological Model) (Collischonn et al., 2007a; Pontes et al., 2017) is a conceptual, semi-distributed, large-scale hydrological model developed for tropical regions. It has been applied in several past forecasting experiments in South American basins, encompassing both deterministic (e.g., Collischonn et al., 2005, 2007b; Fan et al., 2017) and ensemble approaches (Fan et al., 2014, 2016a, 2016b, 2015; Meller et al., 2014; Paiva et al., 2012; Siqueira et al., 2016). Here we used the South America MGB model version presented in Siqueira et al. (2018), hereafter named as MGB–SA. The model uses a vector-based river network divided in reaches of constant length (Δx = 15 km), and basin discretization is obtained by defining the direct catchment area to the individual reaches (i.e., in the form of unit-catchments). Combinations of land use and soil type within each unit-catchment are categorized as Hydrological Response Units (HRU), where water budget and energy balance are computed independently at a daily time step. Propagation of surface, subsurface and groundwater runoff to the main channel is computed with linear reservoirs in order to represent catchment delay and attenuation. Flow routing in the river network and associated discharge, water surface elevation and flood extent are simulated using a 1D local inertia hydrodynamic model based on Bates et al. (2010) with an adaptive time step (few minutes) (Pontes et al., 2017). Hydrodynamic routing has been shown to be important to simulate flood delay in many large rivers of SA (Paiva et al., 2013c; Siqueira et al., 2018).

MGB–SA was manually calibrated with hundreds of in situ observations focusing on a well-balanced simulation, i.e., a reasonable depiction of high and low flows. The model was validated using river water levels, evapotranspiration and terrestrial water storage from multiple data sources, being able to reproduce daily discharges satisfactorily over several regions of South America (70% [55%] of the gauges with KGE [NSE] > 0.6) (Fig. 2). MGB–SA also compared favorably to state-of-the-art global hydrological models with respect to several performance metrics (Siqueira et al., 2018). In this study, no additional modifications were made to the original MGB–SA, and for further details about model setup, parameterization and validation results the reader is referred to Siqueira et al. (2018).

3.2. Meteorological input

3.2.1. Observed rainfall and climate data

The Multi-Source Weighted Ensemble Precipitation (MSWEP) v1.1 (Beck et al., 2017b) was used as the input forcing to initialize forecasts. This is a global-scale dataset (0.25° resolution) that optimally combines precipitation from satellite, reanalysis and in situ gauges. This dataset was chosen because it performed well compared to 22 other satellite-based estimates (Beck et al., 2017c) and was successfully used for MGB–SA calibration (Siqueira et al., 2018). Other meteorological variables needed by the model to compute evapotranspiration (ET) were retrieved from the CRU Global Climate v.2 (New et al., 2002), which provides long-term monthly means (1961–1990) of temperature, pressure, radiation and wind speed for all land areas at 10′ resolution. In order to match these meteorological fields to the irregular computational elements of the hydrological model, data from MSWEP and CRU were interpolated to the unit-catchments of MGB–SA using the inverse-distance-weighted method and nearest neighbor approach, respectively.

3.2.2. Predicted rainfall: ECMWF ensemble data

Precipitation forecasts from the ECMWF Ensemble Prediction System (ECMWF EPS) were acquired between Nov/2010–Oct/2014 through the THORPEX Interactive Grand Global Ensemble (TIGGE) repository (Bougeault et al., 2010). The ECMWF EPS is a 51-member ensemble composed by a control run and 50 perturbed forecasts, produced after perturbing the initial conditions and physical parameterizations of the atmospheric model using the Singular Vectors technique and a stochastic approach (Leutbecher et al., 2017). Forecasts are issued twice a day (00 UTC and 12 UTC) and provided with 6-h accumulated intervals and a forecast horizon up to 15 days ahead. The operational EPS runs with a variable horizontal grid resolution of 32 km up to day 10 and 63 km for days 10–15, but outputs in TIGGE database are freely available to the general public with a 0.5° resolution (it ranges from ~55 km at the equator to ~32 km at the southern tip of SA). The ECMWF EPS has demonstrated relatively good performance in South America with respect to other similar systems (Medina et al., 2019), even when predicted precipitation is propagated to the hydrological forecasts (Fan et al., 2015; Tomasella et al., 2019). Only the 50 ECMWF perturbed members were used in this study.

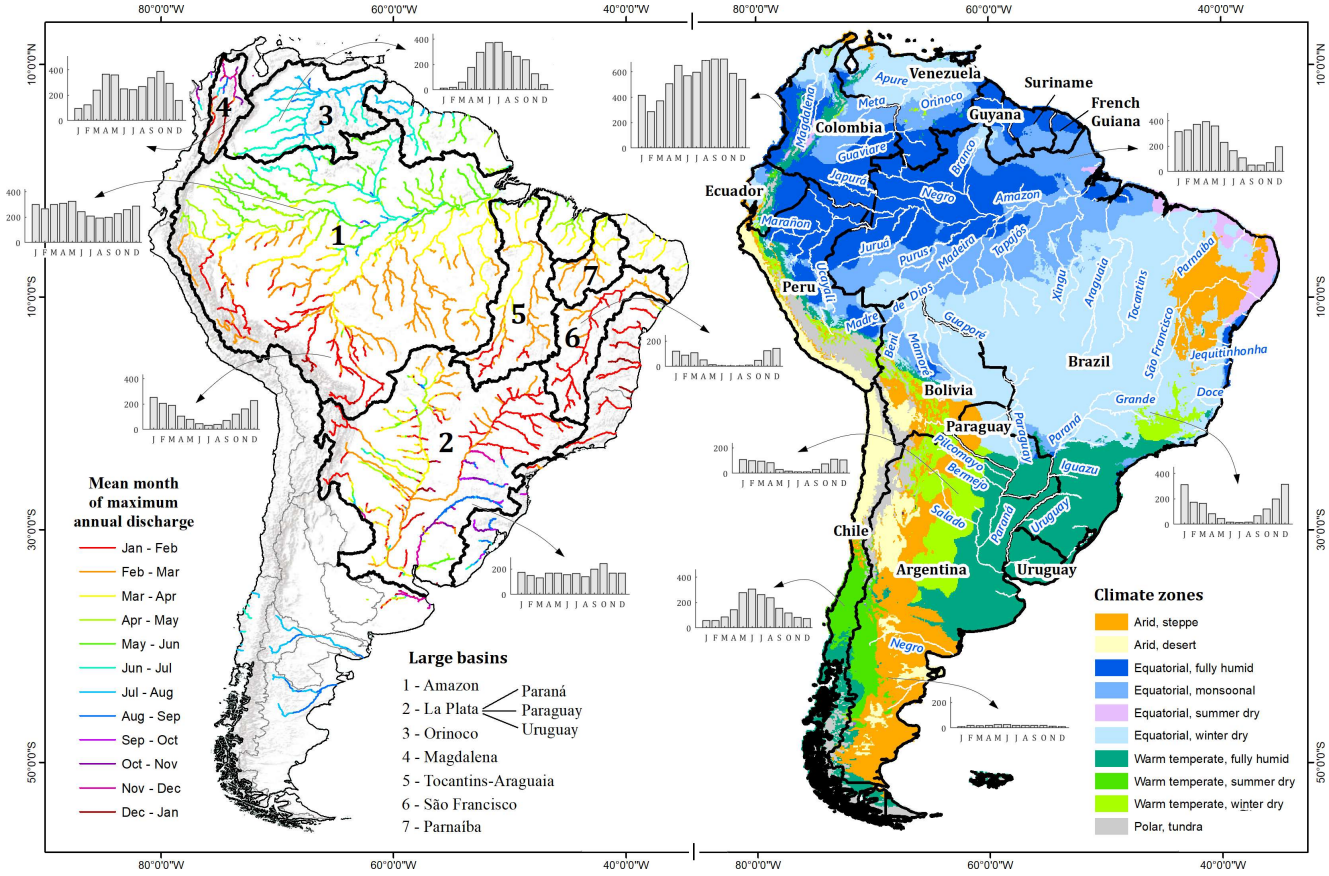


Fig. 1. Hydroclimatic characteristics, political divisions (right) and large basins of South America (left). Histograms represent the annual cycle of precipitation (in mm) derived from the MSWEP dataset (Beck et al., 2017b). The mean month of maximum annual discharge of South American large rivers was computed using the simulated discharge dataset available from Siqueira et al. (2018).

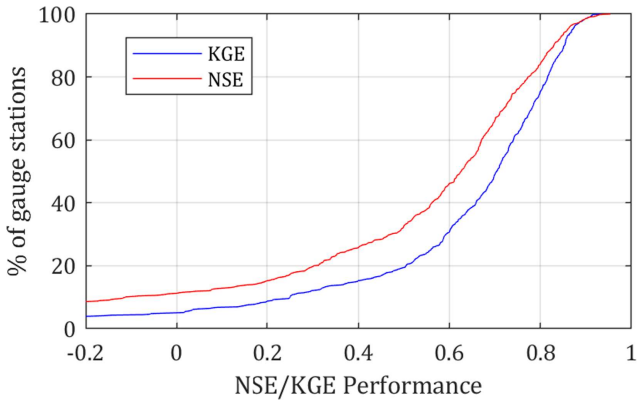


Fig. 2. Cumulative distribution of NSE and KGE performances of MGB–SA in 604 gauge stations across South America, computed for the model calibration period (1990–2010).

3.3. Forecasting setup

3.3.1. Producing hydrological ensemble forecasts

Here, ensemble streamflow forecasts were issued daily by cascading only uncertainties from numerical weather prediction to the hydrological outputs. This was performed within a four-year hindcasting period from 01/11/2010 to 31/10/2014. Although more recent ECMWF forecasts are available in TIGGE repository, we chose to limit our hindcasting period due to computational constraints and to keep coherence with the MSWEP v1.1 precipitation used to calibrate MGB–SA (see Siqueira et al., 2018). MSWEP data from 2015 onwards are available only for more recent versions of this dataset, which include major upgrades (Beck et al., 2019) that would require model recalibration.

To match resolutions of both meteorological and hydrological models, 6-h accumulated precipitation from ECMWF EPS forecasts were temporally aggregated to daily input data and were spatially disaggregated to the centroids of unit-catchments in the same way as done with the observed rainfall. Model state variables were updated by a long-term run using MSWEP data, starting from 01/01/1990 up to each forecast day in order to account for antecedent conditions such as soil moisture and water stored in rivers and floodplains. To produce the ensemble of streamflow forecasts (hereafter named as ECMWF-based forecasts), rainfall from each of the 50 perturbed members of ECMWF EPS was then used to force the MGB-SA model using the updated hydrological states, while other meteorological variables used to compute ET (temperature, wind speed, relative humidity, radiation and pressure) were kept equal to their long-term monthly means along the forecast horizon, similarly to previous approaches in South American basins (e.g., Fan et al., 2014, 2016a, 2015; Meller et al., 2014; Siqueira et al., 2016).

3.3.2. Deriving flow thresholds and proxies of in-situ observations

Early warning systems are often evaluated in terms of exceeding or not a critical threshold level (Bartholmes et al., 2009; Hirpa et al., 2016; Thielen et al., 2009). Although thresholds should be ideally linked to predefined return periods (e.g., 2-yr, 5-yr or 25-yr), floods are relatively rare so that having extreme discharges as a reference may result in capturing a relatively small number of events, especially in short analyzed periods (Bartholmes et al., 2009; Cloke and Pappenberger, 2009). Moreover, information about warning and alert levels might be only

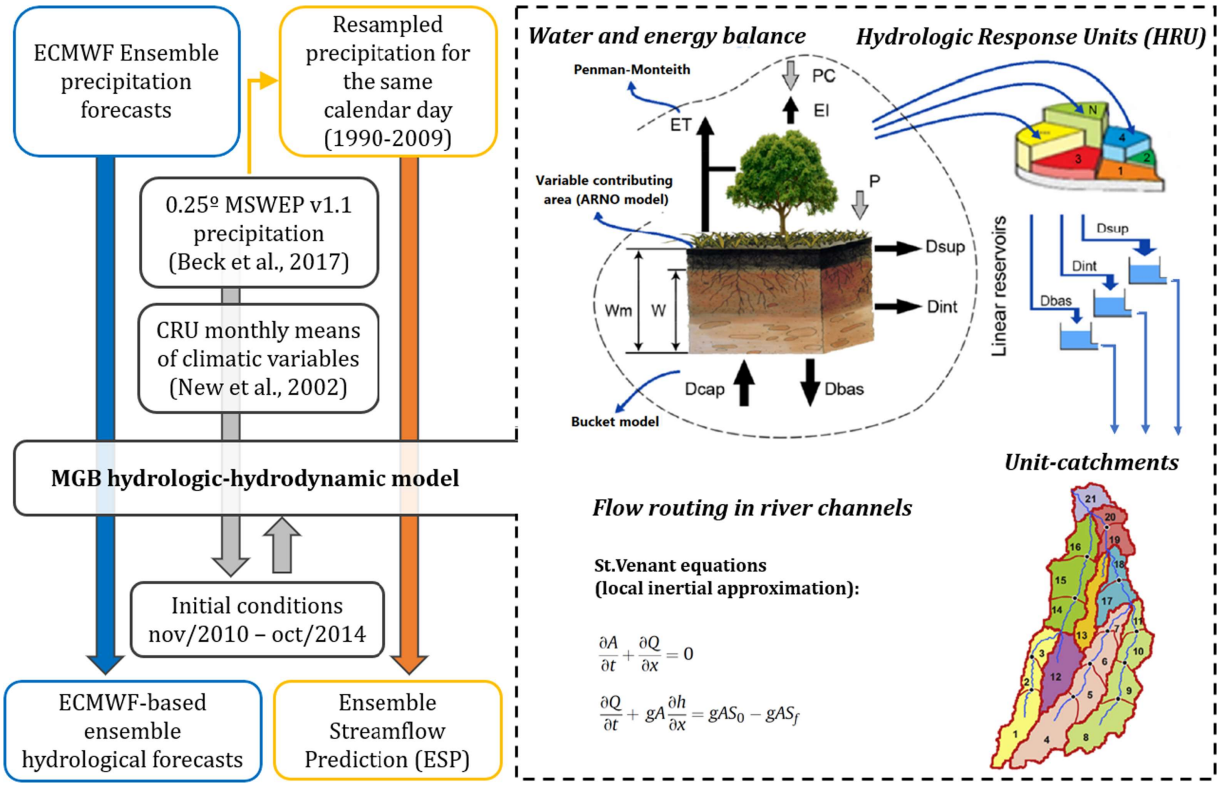


Fig. 3. Scheme of the forecasting setup for producing hydrological ensemble forecasts.

available locally and at specific gauge stations. A feasible solution is to consider simulated discharges as a proxy of in situ observations and to derive thresholds by using predefined quantiles of these proxies (e.g, Bartholmes et al., 2009; Cloke et al., 2017; Olsson and Lindström, 2008; Pappenberger et al., 2011; Thirel et al., 2008; Voisin et al., 2011). The main advantage of this approach is that it does not take into account errors associated with deficiencies in the hydrological model structure, also enabling to evaluate forecast performance over the entire model domain instead of using only a limited number of gauge stations (e.g., Alfieri et al., 2013, 2014; Bischiniotis et al., 2019; Hirpa et al., 2016; Thielen et al., 2009). The above strategy relies on the assumption of a perfect hydrological model, i.e., that errors in streamflow predictions are reflected only by inaccuracies in the predicted precipitation. This is far from true in our case, therefore results derived here have to be seen only as a *potential* forecast skill.

Long-term discharge climatologies (observed proxies) were first extracted for each river location by running the MGB–SA model with MSWEP data from 01/01/1990 to 31/12/2014. To capture a reasonable number of events and conduct a meaningful statistical analysis, reference thresholds equivalent to the 90th percentile (Q_{90}) of non-exceedance flows (i.e., 90% of simulated discharges lying below this value) were derived from the long-term discharge climatology, which is in agreement with other similar studies (e.g., Alfieri et al., 2013; Pappenberger et al., 2011; Thirel et al., 2008).

3.3.3. *Using ESP forecasts for benchmarking*

Hydrological forecasts are often compared with a reference benchmark in order to evaluate the relative performance (the forecast skill), since a similar outcome sometimes can be achieved by a less computationally expensive method (Pappenberger et al., 2015b). The most common examples of benchmarks used for this purpose are the streamflow climatology and the persistence of the last observed discharge, especially for the assessment of flood forecasting systems at continental to global scales (e.g., Alfieri et al., 2013, 2014; Bartholmes et al., 2009; Pappenberger et al., 2011). Selecting the appropriate benchmark is not a straightforward task because benchmarks that are too naïve can easily result in apparent high skill, so the benchmark should be preferably the "hardest to beat" alternative (Pappenberger et al., 2015b). The Ensemble Streamflow Prediction (ESP) technique commonly used to produce or assess the skill of seasonal streamflow forecasts (Arnal et al., 2018; Crochemore et al., 2017; Wood and Lettenmaier, 2006) was selected as the benchmark for this study. Moreover, it has been shown that the ESP (also known as meteorological benchmark) can suit well evaluations of forecasting skill also at the medium-range (Bennett et al., 2014; Pappenberger et al., 2015b). In this approach, an ensemble of meteorological variables (e.g., precipitation) for the same calendar day are retrieved from historical observations and are used to force the hydrological model, considering the state variables updated for the current forecast day.

Taking the first day of the hindcasting period (01/11/2010) as an example, the ensemble of predicted precipitation for the first lead time is obtained by resampling MSWEP data in all years between 1990 and 2009 for the target day (i.e., 01/11/1990, 01/11/1991… 01/11/2009). This is repeated for all lead times and initialization dates. To produce the corresponding ESP forecast, MGB–SA is forced with each one of these precipitation scenarios resampled from past observations, using the updated model states computed for the forecast initialization day (01/11/2010). The resampled precipitation data were extracted from a retrospective period spanning from 01/01/1990 to 31/12/2009, which is independent of the hindcasting one and corresponds to the same calibration period of MGB–SA (Siqueira et al., 2018). Therefore, 20-member ESP forecasts were produced as a benchmark system, similarly to the experiment carried out by Pappenberger et al. (2015b). Although the potential effects of ensemble size were not evaluated in this study, we recognize that using a different number of members regarding ECMWF-based and ESP forecasts may affect the skill scores, as discussed by Ferro et al. (2008). For example, a previous study has shown that reducing the ensemble size from 51 to 20 members led to reductions in CRPSS metric by around 2–3% for lead times up to 5 days (Junk et al., 2015). Fig. 3 summarizes the methods used to produce ensemble

streamflow forecasts.

We reinforce that the term "observations" used for precipitation and discharge in the next sections refer to the MSWEP data and streamflow proxies, respectively.

3.3.4. Forecast assessment

There are many scores and graphical tools to measure the performance of probabilistic forecasts, most of them based on standard methods used in meteorology (e.g., Jolliffe and Stephenson, 2012; Wilks, 2011). For instance, the Continuous Ranked Probability Score (CRPS) (Hersbach, 2000) is a widely-known metric to evaluate the performance of forecasting systems and is often published in official reports (Pappenberger et al., 2015b). CRPS summarizes the overall quality of a probabilistic forecast by comparing its cumulative distribution function (CDF) to a Heaviside step function at the observed value. It is computed as an average of all individual CRPS for a given forecast lead time, according to:

$$\overline{CRPS} = \frac{1}{N}\sum_{i=1}^{N}\int_{-\infty}^{\infty}[F_i(y) - F_i^0(y)]^2 dy \quad (1)$$

$$F^0(y) = \begin{cases} 0, & y \le observed\ \ value \\ 1, & y > observed\ \ value \end{cases} \quad (2)$$

where: $F_i(y)$ is the CDF of the forecast ensemble y, computed for the forecast day i; $F^0(y)$ is the Heaviside step function that equals to 1 when forecast values are greater than the observed value and is zero otherwise; N is the number of forecasts issued.

$\overline{CRPS}$ is finally converted into a skill score as follows:

$$CRPSS = 1 - \frac{\overline{CRPS}_{fcst}}{\overline{CRPS}_{benchmark}} \quad (3)$$

where $\overline{CRPS}_{fcst}$ is the $\overline{CRPS}$ computed for the ECMWF-based forecasts and; $\overline{CRPS}_{benchmark}$ is the $\overline{CRPS}$ computed for the ESP forecasts as the benchmark system. The ECMWF-based system is skillful if CRPSS > 0 and maximum skill occurs when CRPSS $= 1$.

The ability to discriminate between the occurrence and no occurrence of an event is assessed with Relative Operating Characteristic (ROC) curves. Both forecasts and observations are transformed into a set of binary events according to the exceedance or non-exceedance of the reference flow threshold (we used Q_{90} as the threshold, as described earlier in Sect 3.3.2). A 2 × 2 contingency table (Table 1) is computed for different forecast probabilities derived from the ensemble, where an event is forecasted if at least X % (i.e., 10%, 20%, …) of the ensemble members lie above the threshold. ROC curves are then constructed by plotting the corresponding pairs of hit (HR) and false alarm (FAR) rates obtained from the contingency tables:

$$HR_j = \frac{a_j}{a_j + c_j} \quad (4)$$

$$FAR_j = \frac{b_j}{b_j + d_j} \quad (5)$$

where j is the index that denotes the forecast probability.

The tradeoff between hit rate and false alarms can be summarized into a single scalar score given by the area under the ROC curve (AROC). A perfect forecasting system would have an area equal to unity. Using a similar approach to Emerton et al. (2019) we computed the dAROC as a measure of skill, which is defined as the difference between the AROC of ECMWF-based forecasts and the AROC of the ESP benchmark:

$$dAROC = AROC_{fcst} - AROC_{benchmark} \quad (6)$$

Reliability diagrams are diagnostic tools used to evaluate whether the forecast is reliable. A perfectly reliable system would produce forecast probabilities X_i that are consistent with the corresponding frequencies Y_i at which the event (e.g., a threshold being exceeded) is observed. Such diagrams are constructed by plotting X against Y, with the forecast probability values often grouped into bins of equal width (e.g., 10–20%, 20–30%, …) (Bröcker and Smith, 2007). The reliability diagram is conditioned on the forecasts; therefore, the observed frequency for a given forecast probability bin (X_k) and lead time is computed according to:

$$X_k = \frac{O_k}{O_k + NO_k} \quad (7)$$

where O_k is the number of times the event has occurred for the probability bin k (observed proxy exceeding the reference flow threshold Q_{90}); and NO_k is the number of times the event has not occurred for the probability bin k (observed proxy not exceeding the reference flow threshold Q_{90}).

If precipitation forecasts persistently over- or underestimate the rainfall used to initialize the hydrological model, discharge forecasts and consequently the prediction of flood alerts will be consequently biased, which may lead to over- or under-prediction of flood alerts (Alfieri et al., 2014). As large precipitation volumes accumulated over several days are usually the main trigger of floods in large basins, we computed the average difference between the accumulated ensemble mean and the accumulated observed rainfall in order to assess the forecast bias:

$$Bias = \frac{1}{N}\sum_{i=1}^{N}[P_{fcst} - P_{obs}] \quad (8)$$

where: P_{fcst} and P_{obs} are, respectively, the ensemble mean of the accumulated precipitation forecast and the accumulated observed rainfall for the lead time in analysis and forecast day i; N is the number of forecasts issued. Bias in precipitation was computed for accumulation periods of 5 and 15 days.

We first performed an analysis of the predicted precipitation before assessing the hydrological forecasts. Rainfall metrics were computed after averaging pixels of ECMWF (0.5°) and MSWEP (0.25°) in grids of 1.0° × 1.0° to minimize position errors between the observed and predicted rainfall fields. Regarding hydrological forecasts, as we have a particular focus on flood prediction, the computation of skill (CRPSS and dAROC) was carried out only for moderate to high flows. This was performed by selecting only the verification pairs (forecasts/observations) for which the observed discharge exceeds the median of flows (Q_{50}) in the hindcasting period (i.e., the subset of flows < Q_{50} was not included in the computation of $\overline{CRPS}$ and contingency tables). We reinforce that Q_{50} differs from the reference threshold Q_{90} used for ROC curves since the former is used to select a subset from the entire verification data. Table 2 summarizes the forecasting setup and the verification metrics adopted in this study.

Table 1
Contingency table for a given forecast probability and lead time. Forecasts and observations are considered as binary events according to the exceedance or non-exceedance of the reference flow threshold Q_{90}.

		Observed	
		Yes	No
Forecast	Yes	a (Hit)	b (False alarm)
	No	c (Miss)	d (Correct negative)

4. Results

4.1. Analysis of predicted rainfall

Figs. 4 and 5 show the performance maps of precipitation in terms of CRPSS and cumulative bias, respectively. Results are presented separately according to the season of the year (DJF, MAM, JJA and SON)

Table 2
Summary of the forecasting setup and metrics/diagrams used for the assessment of forecasts.

Information	Evaluated system	Benchmark
Precipitation ensemble	Daily precipitation forecasts from the 0.5° resolution, ECMWF EPS	Daily resampled MSWEP precipitation from the past (1990–2009), for the same calendar day
Hydrological model	Hydrologic–hydrodynamic model (MGB–SA): *ECMWF-based forecasts*	Hydrologic–hydrodynamic model (MGB–SA): *ESP forecasts*
Number of ensemble members	50	20
Forecast horizon	15 days	15 days
Verification scores / diagrams	Discharge and Precipitation: CRPSS*	
	Discharge only: dAROC* and Reliability diagrams. The reference flow threshold is the Q_{90} (90th percentile of non-exceedance flows)	
	Precipitation only: Bias	
Hindcasting period	Nov/2010–Oct/2014 (4 years)	
Discharge Observations	Observed proxies: Simulated discharges of MGB–SA using daily MSWEP precipitation data as the model forcing.	

*For discharge, the computation of skill scores was performed only for moderate to high flows, i.e., using the subset of verification data for which observed proxies > Q_{50}.

and for the lead times of 5 and 15 days. Large basins and hydrological regions of South America are also displayed to aid visualization. Areas shown in blue in Fig. 4 indicate skillful precipitation forecasts, whereas gray and red areas demonstrate no added skill and lower overall performance than the reference, respectively. As observed, ensemble precipitation forecasts from the ECMWF have skill over climatology in many regions of SA. The skill is higher at shorter leads (i.e., 5 days) but decreases along the forecast horizon, resulting in values predominantly close to 0 after 15 days. Higher values of CRPSS are mainly observed at eastern and southeastern SA regions, while in the subtropics and mid-latitudes north of 50° S positive CRPSS values can be found all year round, such as in the south of La Plata basin and central Chile. On the other hand, in large areas located at central SA (e.g., Amazon, Tocantins–Araguaia, north of La Plata basin) and also north of the equator like the Orinoco basin (Venezuela), the CRPSS values are notably lower in the wet than the dry season. In most of these areas ECMWF forecasts show little or no added skill even at shorter lead times (5 days), except for the dry season. Interestingly, negative values of CRPSS are markedly seen along a narrow stripe at the western part of the continent (close to the Andes Cordillera), thus indicating a lower performance of predicted precipitation in comparison to daily rainfall climatology in such regions.

The mean cumulative bias in Fig. 5 also exhibits spatial and seasonal variations. ECMWF overestimates precipitation in most of La Plata basin, but mainly in the southwestern Amazon and northwestern SA with bias generally ranging from 10 to 20 mm for 5-day accumulated rainfall (reaching > 30 mm in the case of 15 days). Overestimation in some areas near the tropical Andes and central Orinoco is expressive even for shorter lead times (> 30 mm for 5-day accumulated rainfall), although such regions have abundant precipitation along the year. In turn, predicted precipitation is underestimated in northern SA, central part of Tocantins–Araguaia basin (and to some extent in the Parnaíba

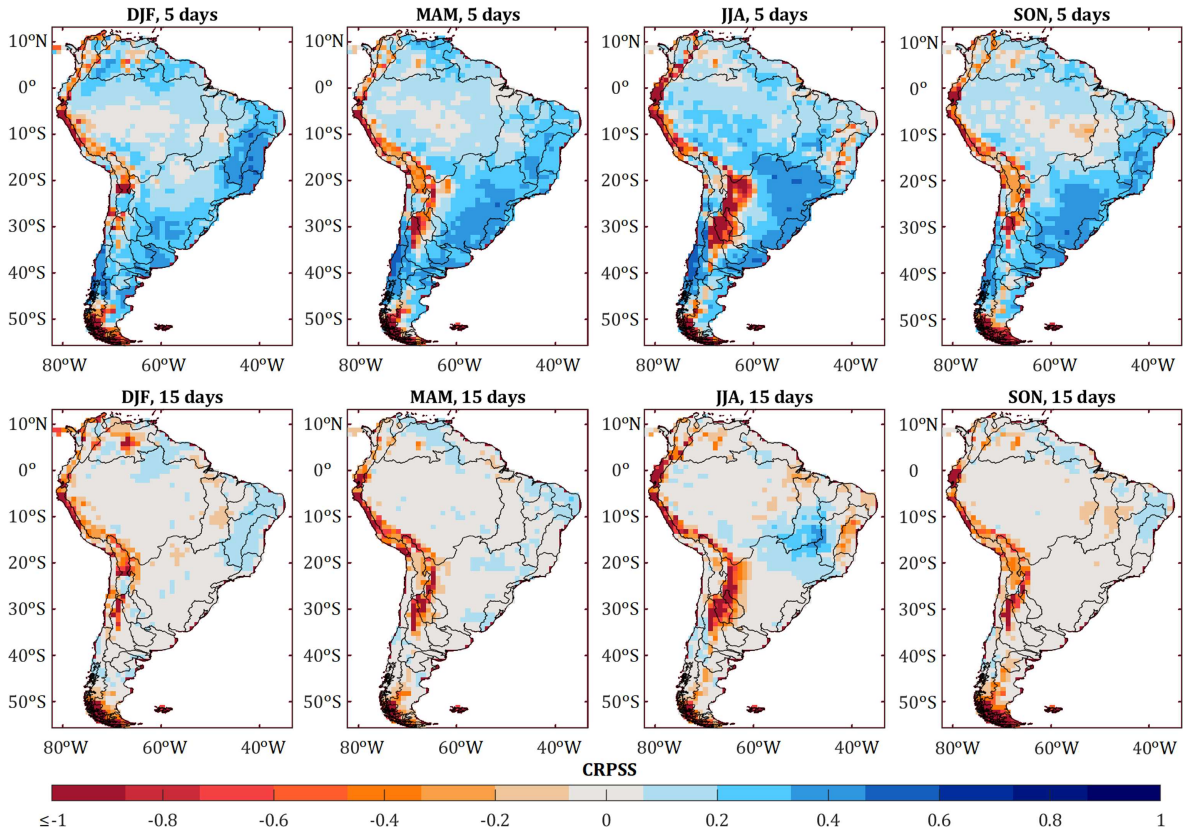


Fig. 4. CRPSS maps of ECMWF predicted precipitation for lead times of 5 (top) and 15 days (bottom). Results are computed for each season of the year, using the resampled daily climatology from MSWEP as the reference benchmark.

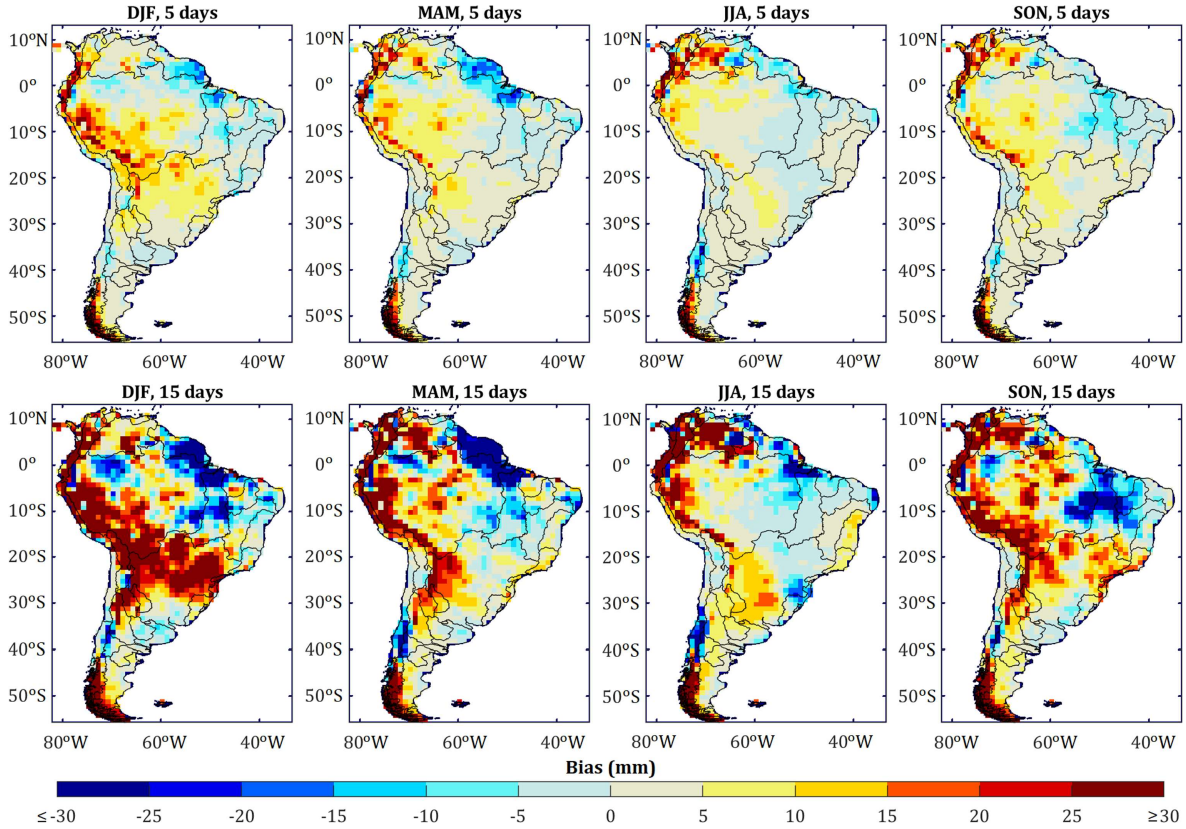


Fig. 5. Maps of mean cumulative bias (in mm) of ECMWF predicted precipitation in comparison with MSWEP data, for accumulation periods of 5 (top) and 15 days ahead (bottom). Results are computed for the ensemble mean and are separated for each season of the year.

basin), as well as in Chile between latitudes of 30° S and 40° S ($-20 < \text{bias} < -10$ mm for 5-day accumulated rainfall). As expected, bias is notably higher in the wet season peak (e.g., Orinoco in JJA, Northern SA in MAM, La Plata in DJF), while in the Tocantins–Araguaia it also perceptible in the onset of rainy season (SON). In southeastern La Plata (Uruguay basin), precipitation is overestimated in the summer (DJF) and underestimated in the winter (JJA) for longer lead times (15 days).

4.2. Assessment of hydrological ensemble forecasts

Performance maps of CRPSS and dAROC for ensemble streamflow forecasts are shown in Figs. 6 and 7, respectively. Metrics were computed for each river segment within the model domain, albeit results are shown only for those with upstream drainage area larger than 10,000 km^2 for visualization purposes. CRPSS is positive in central SA including the north of the Amazon, and increases at the eastern and southeastern part of the continent. ECMWF-based forecasts generally exhibit higher skill at lead time of 5 days, but are still skillful 15 days in advance especially for large rivers. In turn, negative values of CRPSS occur systematically in areas such as central Orinoco, southwest of the Amazon and west of La Plata basins. Except for Orinoco, the lack of skill is observed next to the headwaters at shorter leads and propagates downstream along the forecast horizon. Regarding dAROC (Fig. 7), skill is substantially higher in both southern and eastern Brazil than in other regions. Some rivers draining the Andes have positive dAROC values up to around 10 days, showing a contrasting pattern to CRPSS. Conversely, a considerable fraction of large rivers shows no or little dAROC skill for ECMWF-based forecasts in comparison with the ESP ones. This is broadly observed in the Amazon, but also in the main stem of Orinoco, Araguaia, Paraguay and lower Paraná basins.

To provide a more detailed, regional analysis about the performance of ECMWF-based forecasts, both metrics (CRPSS and dAROC) were plotted against the forecast lead time and basin size for a selection of large basins of the continent (Figs. 8 and 9, respectively). Median values of CRPSS (Fig. 8) are close to 0 (no skill) in basins such as Magdalena, Orinoco, Madeira and Paraguay. A large dispersion of forecast performance is observed in these cases. While the upper 5th percentile (p_5) line is mostly limited to a skill around +0.2, the 95th percentile (p_{95}) reaches values below –1.0 for lead times between 5 and 10 days in advance. Especially for Magdalena, Orinoco and Madeira, there is no clear pattern that relates forecast added skill, drainage area and lead time. Indeed, the signal of skill (positive or negative) is not evenly distributed at the tributaries, which can be visually confirmed in the spatial analysis (Fig. 6). On the other hand, the median of CRPSS is positive in all lead times in Upper Paraná, Uruguay, Parnaíba, São Francisco and Tocantins-Araguaia. Maximum CRPSS (median) is achieved at lead times around 3–5 days because part of the streamflow forecast skill at shorter lead times comes from observed precipitation, reducing to median values close to +0.2 at the end of forecast horizon. Only in the latter basins, skill is higher for larger drainage areas, reaching values slightly over +0.5 in the best case (São Francisco).

According to Fig. 9, a relatively symmetric dispersion of dAROC occurs around the no skill line in large basins located in the northern hemisphere (Magdalena and Orinoco), with most of the values ranging between –0.2 and +0.2. In the other cases, the p_5 boundary is close to 0 for almost all lead times, which indicates that ECMWF-based forecasts usually tend to be at least as skillful as the ESP regarding the detection of high flow events. For the Upper Paraná, Parnaíba, São Francisco and Paraguay, dAROC skill is better for smaller drainage areas than larger ones, thus exhibiting an inverted pattern in relation to CRPSS. In the Uruguay, values of dAROC are higher (lower) for smaller (larger) drainage areas at shorter lead times but start to reduce (increase) along the forecast horizon, resulting in an inverted pattern around 5–10 days in advance.

The boxplots in Fig. 10 summarize the performance metrics (CRPSS

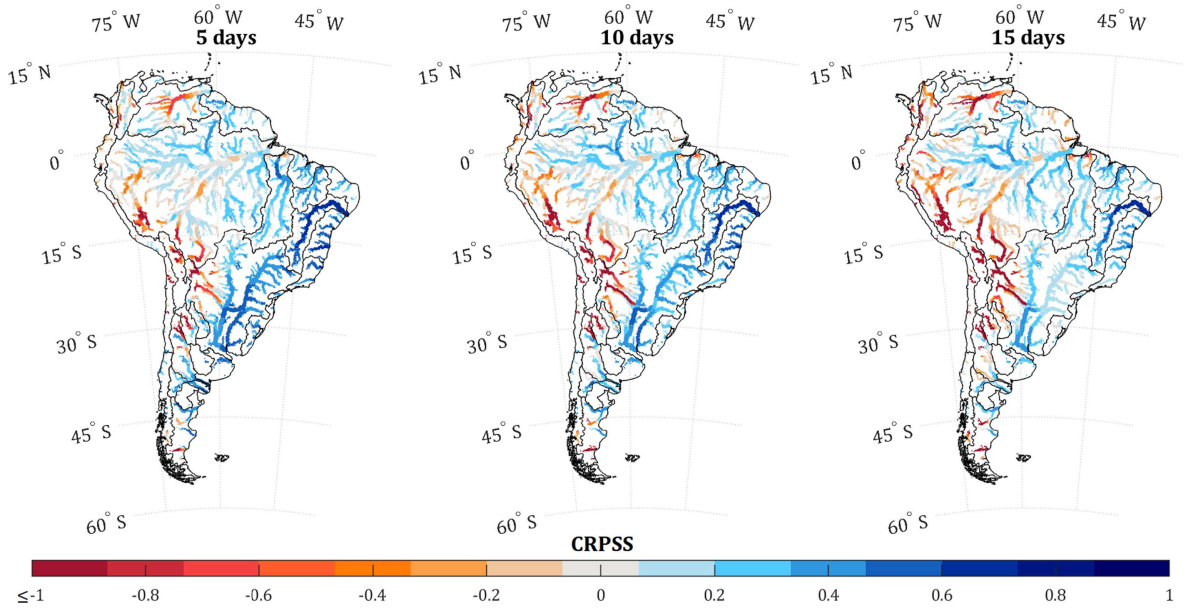


Fig. 6. CRPSS maps of ECMWF-based streamflow forecasts for lead times of 5, 10 and 15 days, using ESP as the reference benchmark. CRPSS is shown only for rivers with drainage area > 10,000 km^2.

and dAROC) in terms of climatic characteristics, considering a fixed lead time of 5 days. The locations of each unit-catchment centroid were categorized according to its corresponding climate region, and results are shown only for drainage areas < 50,000 km^2 to reduce the influence of large rivers that potentially cross multiple climates. Desert and polar regions, which are both concentrated at the western part of South America (Fig. 1), were not included in the analysis as the model does not account for snowmelt processes that may be important for these areas (Siqueira et al., 2018). As observed, interquartile ranges of CRPSS are notably above the no skill line in equatorial with summer or winter dry climates, as well as in warm temperate with fully humid and dry summer regions. For dAROC, skill tends to be higher along areas characterized by a fully humid, warm temperate climate but also in arid (steppe) zones, where the variability of performance is also larger.

To assess the conditional biases of the ECMWF-based forecasts, the reliability diagrams for selected basins and different lead times are shown in Fig. 11. Probabilities of exceeding the Q_{90} threshold are given in bins of deciles (e.g., 10–20%, 20–30%, …) and were plotted in the arithmetic center of the bins. Reliability diagrams were constructed by pooling the ensemble forecasts over the entire basins to increase the sample size in each bin, and the number of times each probability was predicted is shown in the bar charts at the upper left corner of diagrams. In general, the reliability curve expressed by the points over the diagrams tends to fall below the diagonal line. This pattern can be clearly seen in the Magdalena, Orinoco, Paraguay and Madeira basins, suggesting that the ECMWF-based forecasts highly overpredict the probability of exceedance of the reference threshold in these cases. For instance, events for which forecasts are predicting ~80% of chance to occur would be observed around 40% of the times (or even less), and this ratio is also noted in intermediary probabilities. In turn, the

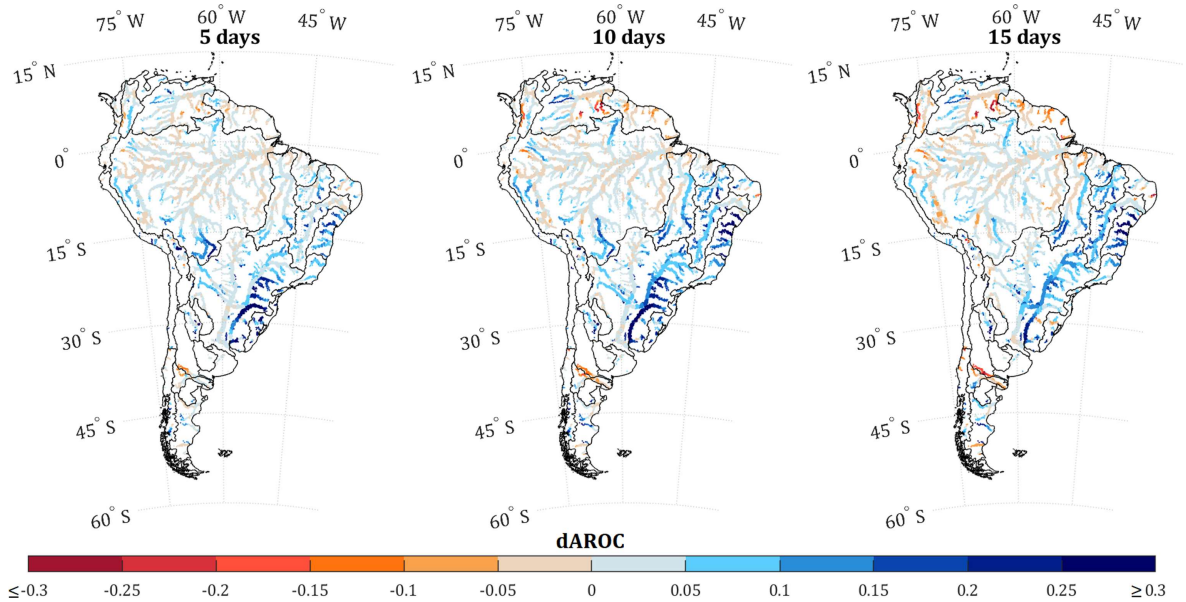


Fig. 7. dAROC maps for ECMWF-based streamflow forecasts for lead times of 5, 10 and 15 days, using ESP as the reference benchmark. Discrimination is computed using the Q_{90} of non-exceedance flows as the reference threshold and is shown only for rivers with drainage area > 10,000 km^2.

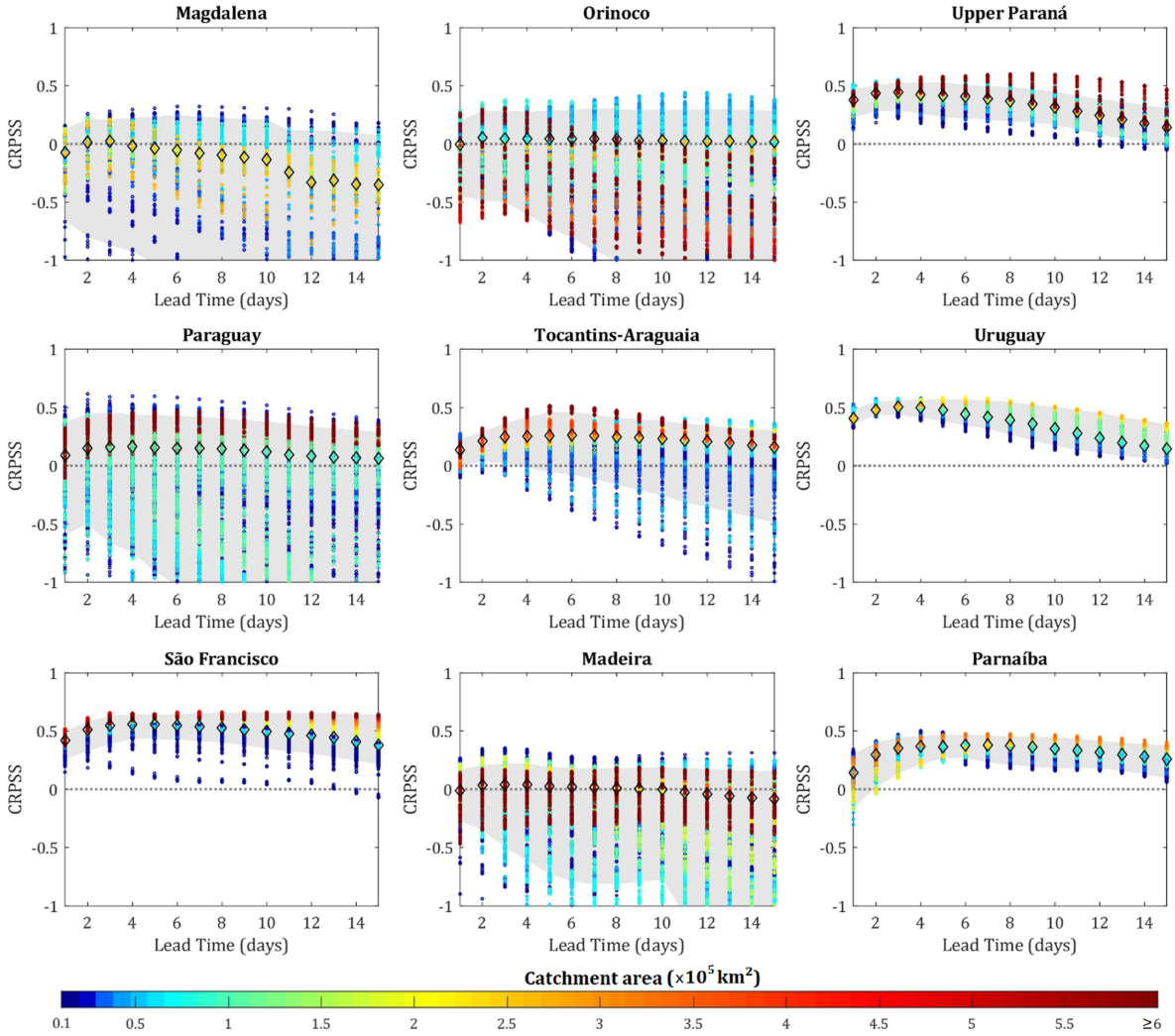


Fig. 8. Relationship between CRPSS, lead time and basin size for different river basins of South America. Colors indicate the drainage area of MGB–SA river segments within each basin. The light gray area represents the interval between the 5th and 95th percentile, whereas the diamond symbols refer to the median of CRPSS values. The dark gray dotted line at 0 corresponds to the no skill threshold.

consistency between forecast probabilities and observed frequencies is better at the scale of other basins, especially in the São Francisco, Tocantins-Araguaia and Parnaíba. A change in the pattern of reliability diagrams over different lead times is generally unnoticeable, being more visible in the highest probabilities and, in particular, in the Uruguay basin for 15 days in advance. The shape of the histograms (note the log scale of y-axis) indicates that ECMWF-based forecasts tend to predict very low probabilities of exceedance of the Q_{90}, albeit a large number of predictions demonstrate probabilities near 100%. This means that the ensemble streamflow forecasts are relatively sharp for the analyzed threshold.

4.3. Event-based analysis

In this section we conduct a visual analysis of selected flood events at Uruguay, Iguazu, Doce, Magdalena and Beni rivers using sequences of ensemble streamflow forecasts (Fig. 12). These were the largest floods within the hindcasting period (Nov/2010–Oct/2014) for the corresponding rivers, and were selected based on relevance or available documentation in the scientific literature. Table 3 provides additional details for the selected locations and their respective flood events according to in situ observations. Critical thresholds (5- and 25-yr return period) used as a reference for flow magnitudes in Fig. 12 were calculated using Gumbel distribution considering the years 1990–2014 to keep coherence between simulated and observed data. As the ECMWF-based forecasts (gray lines) and ESP (not shown) share the same initial conditions of the model reference run (blue line), it is normal that they follow the same path along forecast dates, which sometimes differ considerably from in situ observed discharges (black line).

The selected flood event of the Uruguay river occurred in July 2014, with the peak reaching around 23,000 m^3s^{-1} between return periods of 5 and 25-yrs. ECMWF-based forecasts were successful in predicting the rising, magnitude and also the flood recession for this particular event. Another flood, in the Iguazu river, reached a maximum daily flow of ~18,000 m^3s^{-1} at Salto Caxias dam (57,100 km^2) in June 2014, and was classified as one of the most catastrophic floods after 1970 in upstream locations (Steffen and Gomes, 2018). High discharges in Iguazu river were predicted with increasingly higher probabilities along the forecast dates, and both peak timing and magnitude were relatively well predicted at shorter lead times.

In Doce river (Southeastern Brazil) at Colatina (77,000 km^2), forecasts are shown for the December 2013 flood event (peak of ~9200 m^3s^{-1}). It was the second-most severe flood in the period of 1939–2014 that occurred in the basin, affecting more than 54,000 people and causing 45 fatalities (Tomasella et al., 2019 and references therein). As noted, the simulated discharges for this event were largely underestimated by the MGB–SA forced with MSWEP data. Some ensemble members predicted flows near the simulated 25-yr threshold only at

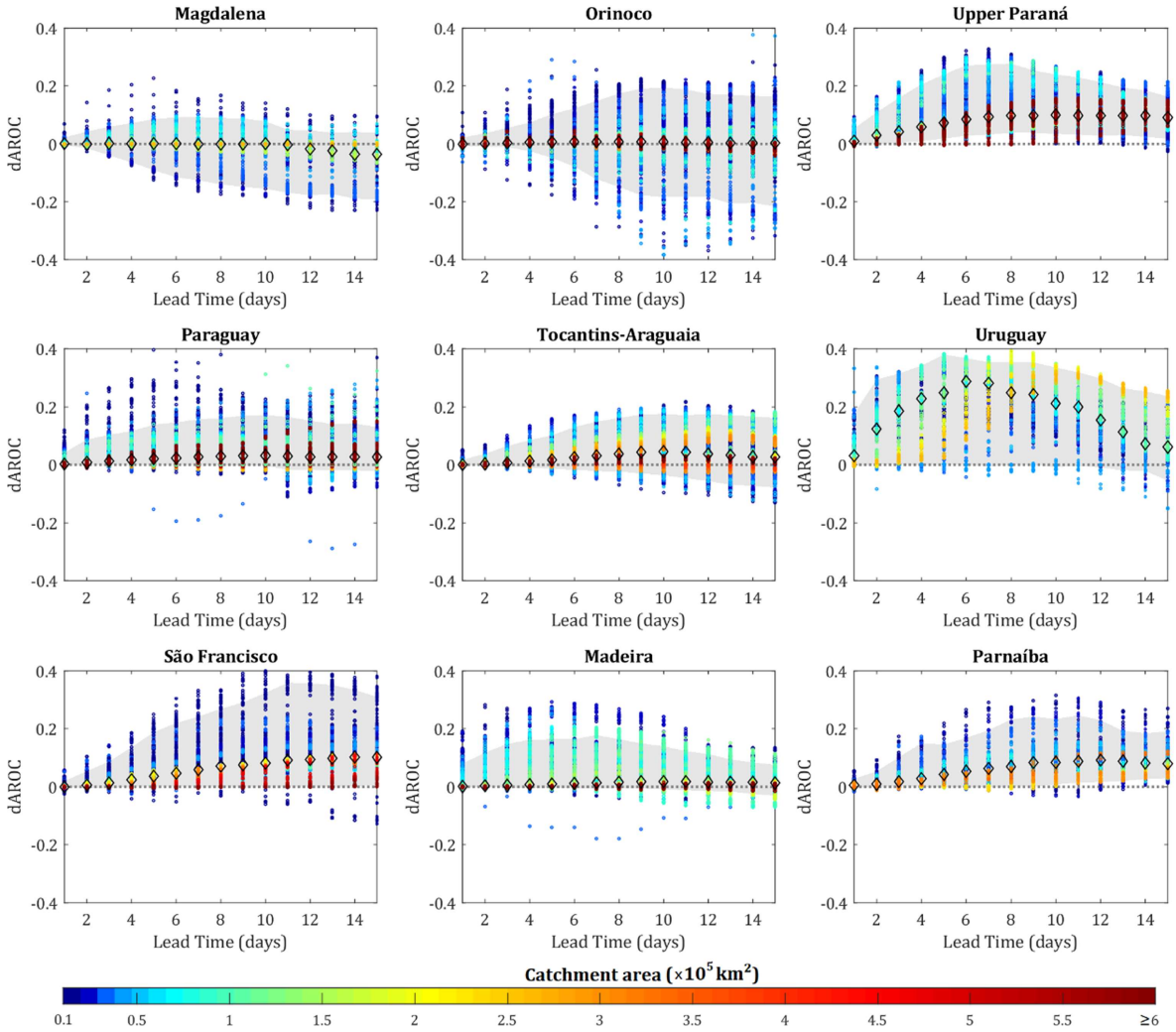


Fig. 9. Relationship between dAROC, lead time and basin size for different river basins of South America. Colors indicate the drainage area of MGB–SA river segments within each basin. The light gray area represents the interval between the 5th and 95th percentile, whereas the diamond symbols refer to the median of dAROC values. The dark gray dotted line at 0 corresponds to the no skill threshold.

longer lead times (for instance, the forecast issued at 12/12/2013), but probabilities of very high discharges reduced for forecast dates closer to the second peak date. For the Magdalena river (Colombia), streamflow forecasts are issued at Arrancaplumas station (56,400 km^2) and refer to a flood that occurred during the extreme 2010–2011 La Niña. That year was characterized by multiple hydrometeorological events (floods, landslides, windstorms, etc) that caused losses of more than US$ 7.8 billion (Hoyos et al., 2013). According to Fig. 12, gradually increasing discharges in Magdalena river have exceeded the 5-yr reference threshold over many days, thus producing a relatively long-lasting flood. ECMWF-based forecasts predicted the streamflow increase in a persistent manner several days in advance, however with considerable ensemble spread.

Finally, forecasts in the Beni river (Bolivia) at Rurrenabaque (68,900 km^2) are shown for the extreme January 2014 floods, which were dramatic for both local and downstream Amazonian communities. According to Espinoza et al. (2014), around 68,000 families and more than 40,000 ha of agriculture were severely affected by the flood event. It was characterized by two main peaks with around 10 days of difference, with discharges above 20,000 m^{3}s^{-1} in both cases. Streamflow forecasts were marked by a very large ensemble spread in all dates and lead times, thus reflecting a very low predictability in this location. Although the simulated discharges were not in agreement with the observed ones, very high discharges exceeding the simulated 25-yr threshold were sequentially forecasted by a considerable number of ensemble members, which indicated high probabilities of extreme events during the analyzed period.

5. Discussion and limitations

The analysis carried out in the previous section suggests that medium-range (up to 15 days) ensemble streamflow forecasts based on the ECMWF EPS are skillful over many regions of South America, when compared to ESP forecasts that rely on resampled rainfall from historical observations as the model forcing. The assessment was performed in a 4-year period between Nov/2010 and Oct/2014 using a continental-scale hydrologic-hydrodynamic to produce hydrological forecasts, which was specifically developed for large-scale hydrological analysis in this continent (Siqueira et al., 2018).

We first compared the precipitation forecasts from the ECMWF EPS to the resampled rainfall predictions that were both used to force the hydrological model. The ECMWF EPS performs well in the east side and southern regions of SA, but has little or no added skill over climatology in the central-north part of the continent and particularly along the Andes. By comparing ECMWF EPS and GEFS precipitation forecasts in Brazilian territory, Medina et al. (2019) also highlighted the former as being skillful at east and southern regions, with a lack of skill in the Amazon basin. Indeed, the forecast skill of atmospheric models is

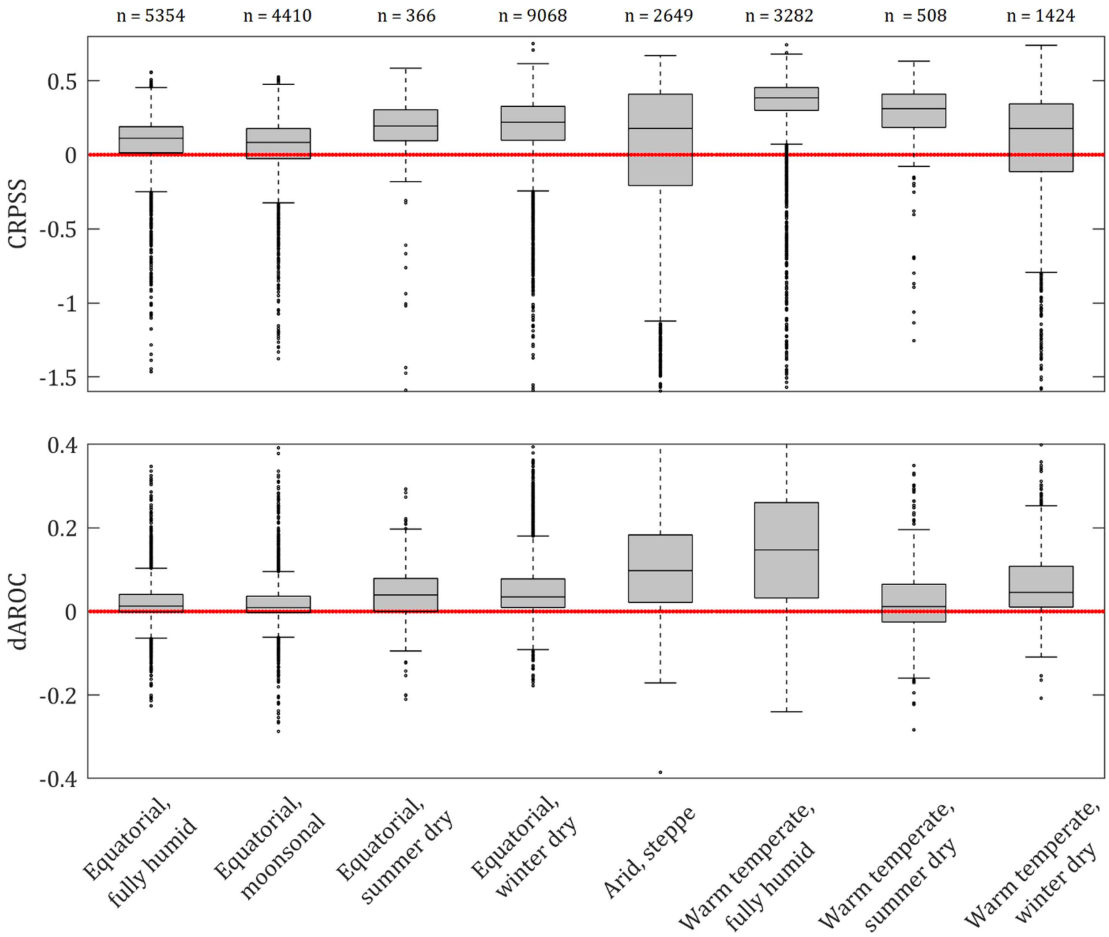


Fig. 10. Skill scores for different climate types in South America considering a lead time of 5 days. Values at the top indicate the number of unit-catchments inside each boxplot. Results are shown for drainage areas less than 50,000 km^2.

known to be lower at the tropics than midlatitudes (Dias et al., 2018; Zhu et al., 2014), since rainfall in the former mostly occurs through deep convection activity that is highly parameterized and one of the major sources of uncertainty in NWPs. In addition, the low CRPSS in areas close to the Andes Cordillera comes together with a positive bias, which was observed systematically over all seasons. Kidd et al. (2013) compared predictions from the ECMWF model to TRMM rainfall data and found a similar pattern of overestimation over the Andes, especially during the austral summer. Janowiak et al. (2010) also found a substantial overestimation in both north and central parts of SA, when assessing forecasts from the ECMWF model against GPCP and CMORPH rainfall estimates. Results are also in agreement with those from Bischiniotis et al. (2019), who evaluated the performance of GloFAS in Peru and stated that ECMWF EPS tends to forecast higher precipitation than the forcing (ERA-Interim reanalysis) used to initialize the hydrological model.

Notwithstanding, there are important remarks on the MSWEP data used as the observation. The latter is a good option to ensure consistent rainfall fields in both space and time over the continental domain, since the availability of gauge stations greatly varies from one region to another. In turn, MSWEP largely depends on the anomalies derived from satellite rainfall estimates over South America (Beck et al., 2017b), for which the associated shortcomings are documented for many countries of the continent (e.g., Blacutt et al., 2015; Dinku et al., 2010; Mantas et al., 2015; Salio et al., 2015). Particularly in the Andes, where the largest discrepancies between MSWEP and ECMWF EPS were found, the spatial variability of precipitation can be quite impressive. Interesting examples are the rainfall hot spots in the Bolivian and Peruvian foothills where annual gradients can reach up to 190 mm km^{-1} (Espinoza et al., 2015), which are certainly misrepresented in the coarse resolution precipitation grids. A common drawback of satellite products in this region is that they typically overestimate light precipitation and underestimate high precipitation rates (Blacutt et al., 2015; Mantas et al., 2015; Salio et al., 2015; Zambrano-Bigiarini et al., 2017). Besides, the detection accuracy of satellites has been found to deteriorate in altitudes between 500 and 2000 m as a potential effect of the orographic uplift of moist air moving towards the Andean slopes (Dinku et al., 2010; Mantas et al., 2015), often related to warm rain processes for which the infrared threshold (of cloud top temperature) may be too low to detect precipitation (Dinku et al., 2010; Hobouchian et al., 2017; Satgé et al., 2016). Heavy rainfall that develops in such mountainous regions may also lack a vertical structure with densely populated frozen hydrometeors, resulting in weak scattering signals for microwave retrievals (Hobouchian et al., 2017). Large areas in the central Andes, like the Altiplano in Bolivia and northern Chile, are very arid and sometimes located in very high altitudes (> 3500 m), and may be affected by subcloud evaporation and presence of snow-covered regions that can lead to overestimation (Salio et al., 2015; Satgé et al., 2016; Zambrano-Bigiarini et al., 2017). Although these shortcomings make it difficult to evaluate the actual skill of predicted precipitation and related biases, satellite estimates are still expected to be more correlated with in situ data when compared to NWPs (Lu et al., 2010).

As expected, the geographical distribution of CRPSS regarding the streamflow forecasts is akin to that observed in predicted rainfall. On the other hand, while overall skill in precipitation decreases to values close to zero for a 15-day range in almost all regions, streamflow forecast skill is still positive two weeks in advance in many rivers due to the delay produced by the processes that transform rainfall into runoff

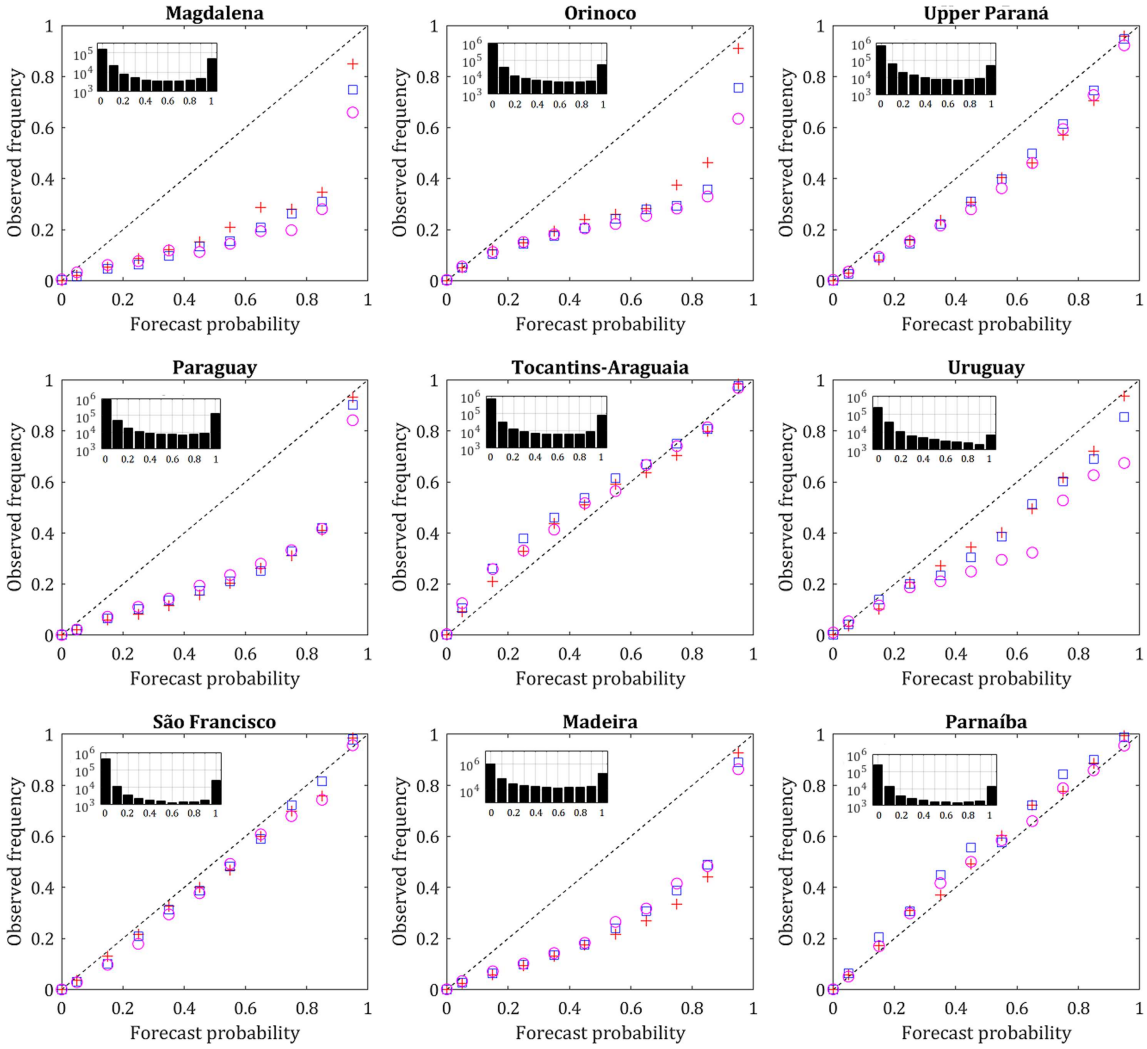


Fig. 11. Reliability diagrams of ECMWF-based streamflow forecasts for different South American basins. Diagrams were constructed by pooling the ensemble forecasts over each basin and are shown for lead times of 5 (red cross), 10 (blue square) and 15 days (magenta circle). The diagonal dashed line represents a condition with perfect reliability. The sharpness histograms show the sample size (log scale) in each bin and are presented for the lead time of 5 days. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and river discharge, moreover in large river basins. This suggests that ECMWF-based forecasts could perform better than ESP even at longer lead times (i.e., sub-seasonal ranges), which may be relevant also for hydropower and water management purposes. This is highlighted for the Upper Paraná basin in the study of Quedi and Fan (2020), who found that ECMWF-based forecasts are skillful when compared to ESP for lead times up to 3–4 weeks ahead.

At the regional scale, the ECMWF-based forecasts are potentially more skillful in basins such as Uruguay, Upper Paraná, São Francisco, Tocantins-Araguaia and Parnaíba, albeit less in Magdalena, Orinoco, Madeira and Paraguay. The former ones showed a similar pattern of overall skill, for which the performance declines for longer lead times and tends to be better for larger drainage areas, similarly to general findings from other studies (e.g., Alfieri et al., 2014; Bischiniotis et al., 2019; Quedi and Fan, 2020). This seems, however, not to be valid for all cases, as such relationship was not found in Magdalena, Orinoco and Madeira basins due to distinct forecast performances at their tributaries. Moreover, forecast probabilities in the latter basins were consistently too large relative to the corresponding conditional event frequencies, which is linked to the positive bias in the predicted rainfall that occurs mainly on the wet season. This overprediction would probably lead to a high number of alerts being issued and, consequently, a high number of false alarms considering the analyzed reference threshold (Q_{90}). One should note that a positive bias in forecast probabilities also increases detection. The area under the ROC curve is more sensitive to hit rate than false alarms (because correct negatives of high flow events occur most of the time) and this could be one of the reasons for which a positive dAROC was found in rivers draining the Andes. It is important, however, to keep in mind the potential issues related to rainfall observations used herein.

The relation between basin size and discrimination skill is also spatially dependent. In large SA basins located north of the equator we found no relationship probably due to biases and low precipitation skill. For some other basins, however, the discrimination skill was found to decrease with drainage area, even for longer lead times. This may be explained by the fact that large rivers are often dominated by initial conditions, that is, the predicted rainfall only becomes important to the streamflow forecasts after a certain number of days. Propagation of floods tends to be very slow in regions with mild slopes that are often regulated by floodplains, as typically observed in large South American basins like the Paraguay (Paz et al., 2011), Amazon (Paiva et al., 2013a), Araguaia (Pontes et al., 2017) and Magdalena (Angarita et al., 2018). For instance, Paiva et al. (2012) clearly show that, for the first weeks (and months in some cases), the relative importance of rainfall forecasts is negligible in comparison with hydrological initial conditions in tributaries at the Central Amazon. Our results provide

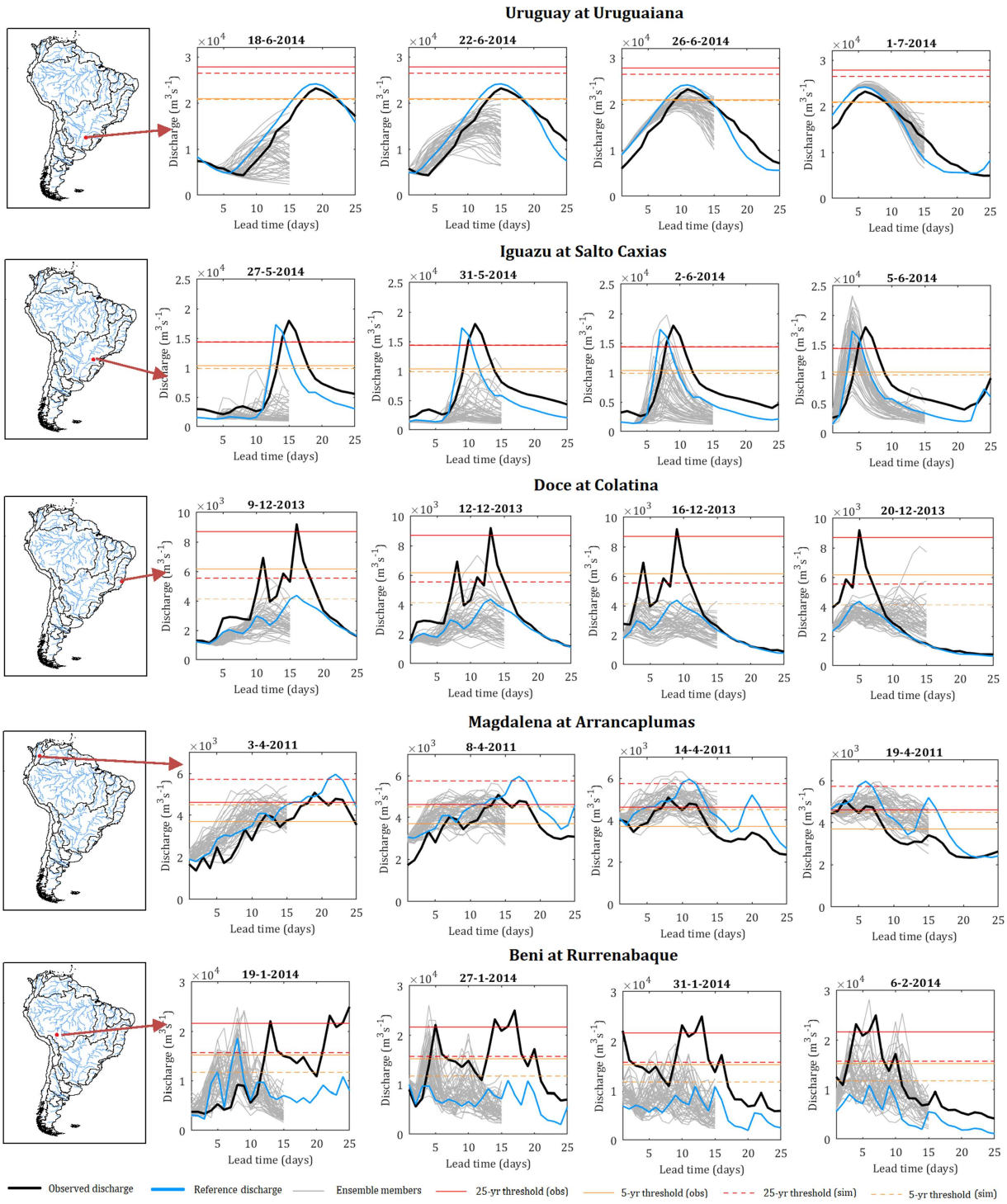


Fig. 12. ECMWF-based streamflow forecasts issued for selected flood events. Black and blue thick lines represent the observed and reference discharge (*proxy*), respectively, whereas gray thin lines represent the forecast ensemble members. Dates at the top of figures indicate the day for which the forecast was issued, starting at lead time 0. Flow thresholds correspond to return periods of 5 (orange) and 25 years (red) and were computed for in situ observations (solid line) and simulated discharge (dashed lines) using Gumbel fit and data from 1990 to 2014. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

additional information to those obtained by Alfieri et al. (2013) when evaluating the GloFAS system, who found that in large rivers of SA (such as the Amazon and Paraná) values of area under ROC curve remain high (> 0.7) over several days in advance. It is worth mentioning that Alfieri et al. (2013) adopted random forecasts (AROC = 0.5) as a reference for discrimination ability. This may limit the assessment of discrimination skill because even ESP forecasts can have high levels of detection in such rivers at shorter lead times, due to the little influence of predicted precipitation with respect to the hydrological initial conditions.

Notably, both overall and discrimination skill had a better performance in the fully-humid, temperate climate, which in turn coincides with southern Brazil, Uruguay and northeast Argentina (southeastern SA). The absence of seasonality potentially makes the ESP benchmark

Table 3
Characteristics of the selected flood events based on in situ observations.

Location	River	Basin	Peak date	Maximum daily observed discharge (m^3s^{-1})
Uruguaiana (190,000 km^2)	Uruguay	Uruguay (La Plata)	06/07/2014	23,190[a]
Salto Caxias dam (57,100 km^2)	Iguazu	Upper Paraná (La Plata)	09/06/2014	17,970 (reservoir naturalized flows)[b]
Colatina (77,000 km^2)	Doce	Doce (Southeast Brazil – Atlantic Coast)	25/12/2013	9196[a]
Arrancaplumas (54,600 km^2)	Magdalena	Magdalena	21/04/2011	5070[c]
Rurrenabaque (68,900 km^2)	Beni	Madeira (Amazon)	double peak, 31/01/2014 (1st), 12/02/2014 (2nd)	22,130[d] (1st), 25,000[d] (2nd)

Data sources: [a]ANA(Brazil), [b]ONS (Brazil), [c]IDEAM (Colombia) and [d]ORE-HyBam (Amazon/Orinoco).

less useful, thus enhancing the relative performance of ECMWF-based forecasts regarding the former. In Australia, Bennett et al. (2014) also found the forecasting system performing better in temperate regions than in other climate types, especially the arid ones with intermittent flows. Moreover, even though severe hydrometeorological events in southeastern SA may be related to MCSs (Demaria et al., 2011; Salio et al., 2007), floods in larger rivers in this region are often triggered by frontal systems during the cold season (Berbery and Barros, 2002; Vera et al., 2002) that can be reasonably well predicted by atmospheric models (Zhu et al., 2014).

The scale of our study indicates to which extent the findings from hydrological forecasting studies in specific regions of SA may be applicable to other parts of the continent. Fan et al. (2015) reported good performance of streamflow forecasts in the Tocantins, Doce (East Brazil) and Upper São Francisco basin when conducting experiments with ECMWF and other precipitation ensembles from TIGGE dataset. Tomasella et al. (2019) found that streamflow forecasts based on the ECMWF EPS were comparable to those obtained from a regional, high-resolution (Eta) EPS for drainage areas larger than 20,000 km^2 in Doce basin, while Siqueira (2015) showed that ECMWF EPS performed similarly to the same Eta model tested by Tomasella et al. (2019) at the short-term range in Southern Brazil. Although these studies focused in comparing the performance of different ensemble forecasting systems rather than skill to a benchmark, one should note that such studies were carried out in locations at east and southeastern South America where the ECMWF EPS was found here to perform reasonably well. However, our results indicate that such forecast performances may not be transposable to the western areas of the continent (e.g., rivers draining the tropical Andes region).

It is worth mentioning again that our results were obtained by assuming simulated discharges as surrogate of observed data (proxies), which were produced by forcing a hydrological model with precipitation estimates from MSWEP. While this is an interesting way to assess which regions ensemble precipitation forecasts might be useful for hydrological forecasting (e.g., Alfieri et al., 2014; Bartholmes et al., 2009), predicted flows are likely to exhibit large deviations from the observed hydrographs. In fact, hydrological models can significantly contribute to the total uncertainty for both short- (Seo et al., 2006; Velázquez et al., 2011; Zappa et al., 2011) and medium forecast ranges (Paiva et al., 2012; Pappenberger et al., 2011; Sharma et al., 2019), and this can be even more important for large-scale hydrological models with simplified calibration (as the MGB–SA) that are forced by other sources than in situ precipitation (Siqueira et al., 2018; Sperna Weiland et al., 2015). Future studies should advance by making use of in situ data where possible or producing better estimates of discharge as a proxy, for instance, through data assimilation methods that have demonstrated promising results in the region (e.g., Paiva et al., 2013b; Wongchuig et al., 2019).

We understand that there are potential impacts of using quite old precipitation forecasts (2010–2014). These are likely to be improved in more recent ECMWF model versions. However, significant improvements are not expected in regions close to the tropical Andes and Amazon basin where results were found here to be relatively poor, given the convective nature of rainfall and its very high spatial variability, as well as the lack of observations that are needed for data assimilation in the atmospheric model. Another limitation is that we did not perform bias correction in ensemble precipitation forecasts. Indeed, bias correction could improve the skill of streamflow forecasts relative to the ESP to some extent, but the known existing uncertainties in the MSWEP product, especially in the west side of SA where the relative bias is high, do not allow to reliably calibrate bias correction techniques in these areas. In addition, bias correction methods usually rely on long and homogeneous time series of both observed and predicted precipitation, while operational forecasts such as the ones from TIGGE repository (as used here) continuously improve over time as atmospheric models are updated. Potential benefits of bias correction strategies with the support of long series of retrospective precipitation forecasts from the same model cycle (i.e., reforecasts) should be investigated in a future work.

The use of a threshold corresponding to the Q_{90} of non-exceedance flows is here justifiable, as the 4-year hindcasting period is relatively short to capture relevant flood events in a single river location. Selecting fixed quantiles from simulated discharges to discriminate between the occurrence and non-occurrence of an event is a common practice at national to global-scale forecasting (e.g., Alfieri et al., 2013, 2014; Bartholmes et al., 2009; Bischiniotis et al., 2019), but are not expected to represent flood conditions equally well in the whole South America. For instance, the Q_{90} threshold may be low in rivers with large variability of flows (e.g., Uruguay), albeit high in others where the flood pulse is very slow and unimodal (e.g., Amazon, Orinoco). Besides, the ECMWF forecasts were shown to be biased compared with the MSWEP data used as the model forcing to derive flow thresholds, which may cause a negative effect in forecast exceedance probabilities. Producing reference thresholds based on a long-term model run with reforecast data rather than observed precipitation (e.g., Hirpa et al., 2016) and defining them according to the lead time in analysis (e.g., Alfieri et al., 2019) may allow further improvements on forecast detection skill.

While the assessment of hydrological forecast skill should be continuously evaluated over time (e.g., Alfieri et al., 2014; Pappenberger et al., 2011), it is also important to outline potential research directions based on recent experiments on large-scale (i.e., continental to global) operational forecasting systems. For example, some studies are currently testing multimodel frameworks (e.g., Kauffeldt et al., 2016) and combining multiple forecasts through postprocessing methods (e.g., Zsoter et al., 2016) in order to better represent the inherent uncertainties. Data assimilation of other variables than in situ discharge such as flood extent maps (e.g., Revilla-Romero et al., 2016) or soil moisture from remote sensing products (e.g., Wanders et al., 2014) may be promising to update state variables and initial conditions of the hydrological model, as streamflow records are often limited in terms of spatial coverage. Furthermore, downscaling and routing the predicted runoff from well-established global forecasting systems (like GloFAS) through a high-resolution river network (e.g., Perez et al., 2016; Snow

et al., 2016) could be interesting to reduce computational costs while improving the local relevance of warnings.

6. Conclusions

This work presented a comprehensive evaluation of the potential skill of medium-range ensemble streamflow forecasts (up to 15 days ahead) for flood prediction in large basins of South America (SA). Raw ensemble precipitation forecasts issued by the ECMWF EPS were used as input to a continental-scale, fully coupled hydrologic–hydrodynamic model to identify regions where the predicted precipitation is expected to produce skillful streamflow forecasts, i.e., after the former is propagated through a perfect hydrological system (namely ECMWF-based forecasts). The assessment of potential skill was carried out having Ensemble Streamflow Prediction (ESP) forecasts as the benchmark, which resamples past rainfall observations (here from the MSWEP dataset) to drive the same hydrological model. Our main findings can be summarized as follows:

- ECMWF-based forecasts are skillful up to 15 days in advance mainly in eastern and southeastern SA. On the other hand, overall skill is considerably lower in large rivers draining the Andean region (at western SA regions), where ECMWF-based forecasts often have lower performance than the ESP;
- ECMWF-based forecasts show skill in discrimination between occurrence and non-occurrence of events compared to the ESP mainly over southern Brazil;
- Both overall and discrimination skill of ECMWF-based forecasts is higher in warm temperate, fully-humid regions than other climate types;
- A global relation between basin size and overall forecast skill was not identified. Although in large basins such as Uruguay, Upper Paraná, São Francisco, Tocantins–Araguaia and Parnaíba the skill of ECMWF-based forecasts reduces with lead time and is generally higher for larger catchment areas compared to the ESP, in others like Magdalena, Orinoco and Madeira the overall skill does not follow the same behavior since it varies geographically along basin tributaries;
- The relation between basin size and discrimination skill is also spatially dependent. In Upper Paraná, Parnaíba, São Francisco and Paraguay basins it is generally higher for smaller drainage areas, whereas for larger basin sizes it tends to increase with lead time as an effect of hydrological initial conditions. On the other hand, in large basins of SA north of the equator no relationship was found;
- Forecast probabilities are highly overestimated in Magdalena, Orinoco, Paraguay and Madeira basins, as an effect of wet biases in predicted precipitation (in relation to MSWEP) that occur mainly during the rainy season.

Our findings also reinforce previous statements that ensemble precipitation forecasts can be valuable to drive continental hydrological models, thus providing large-scale hydrological predictions to complement existing systems or to encompass regions where no forecasts are currently available (Alfieri et al., 2013; Emerton et al., 2016; Thielen et al., 2009; Thiemig et al., 2015). Although the results presented herein offer a broad picture about the potential skill of streamflow forecasts based on meteorological EPS, important technical challenges are posed. Given the low coverage of hydrometeorological gauge stations providing real-time data in the context of SA, the use of satellite estimates (or satellite-based products that merge data from multiple sources) is — and will be — of paramount importance to initialize the hydrological states before precipitation forecasts can be used as model input. The known limitations of such estimates combined to the model errors (Siqueira et al., 2018) undoubtedly lead to significant hydrological uncertainties, and these must be accounted for in order to shed light on the actual skill of the forecasting system. Therefore, there are major opportunities of postprocessing and data assimilation at the continental domain of SA to make the best use of available observations and remote sensing data. Nonetheless, as the topic of ensemble hydrological prediction is still incipient in SA, we expect that our results can also encourage regional forecast experiments in SA locations where hydrological forecasts may benefit from a meteorological EPS (such as the ECMWF).

CRediT authorship contribution statement

Vinícius Alencar Siqueira: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing - original draft, Writing - review & editing. **Fernando Mainardi Fan:** Conceptualization, Writing - review & editing. **Rodrigo Cauduro Dias de Paiva:** Conceptualization, Writing - review & editing. **Maria-Helena Ramos:** Conceptualization, Writing - review & editing. **Walter Collischonn:** Conceptualization, Supervision, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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